

Sérgio Makhambetov

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EDUCATION

ETH Zurich (Swiss Federal Institute of Technology) | *M.S. Computer Science* 2013 - 2015

- Completed thesis on distributed consensus algorithms, achieving 92% optimization in Byzantine fault tolerance simulation
- Led computer science student association with 300+ active members, organizing 8 technical workshops annually

University of Waterloo | *B.A.Sc. Software Engineering* 2009 - 2013

- Graduated with distinction; maintained 3.8/4.0 GPA in core CS courses
- Co-founded university's competitive programming club, growing participation from 15 to 120 members over 3 years

EXPERIENCE

Stripe | *Staff Software Engineer, Payments Platform* 2021 - Present

- Architected distributed payment processing system handling 50M+ transactions daily across 135+ countries, reducing latency by 35% and saving \$2.3M annually in infrastructure costs
- Led team of 6 senior engineers to rebuild core settlement engine, decreasing reconciliation errors by 99.7% and preventing \$18M in customer disputes
- Mentored 12 junior and mid-level engineers, with 8 promoted to senior roles within 18 months; established quarterly technical leadership program

Cloudflare | *Senior Staff Engineer, Edge Network Infrastructure* 2018 - 2021

- Designed and deployed next-generation edge caching layer processing 1.5 trillion requests monthly, improving cache hit ratio from 68% to 91% and reducing customer latency by 42%
- Spearheaded migration of 500+ critical services to new networking stack, eliminating 15 years of technical debt and reducing operational incidents by 78%
- Drove adoption of structured logging across 200 microservices using custom distributed tracing platform, improving MTTR for production issues from 45 minutes to 8 minutes

Notion | *Staff Software Engineer, Real-Time Collaboration* 2016 - 2018

- Built conflict-free replicated data type (CRDT) engine supporting collaborative editing for 2M+ concurrent users, achieving sub-100ms sync latency
- Optimized WebSocket message protocol, reducing bandwidth consumption by 67% and supporting 40% more concurrent connections on same infrastructure
- Designed and implemented observability stack processing 200M events daily, enabling oncall engineers to diagnose 95% of critical issues within first 5 minutes

Palantir Technologies | *Senior Software Engineer, Data Integration Platform* 2014 - 2016

- Led architectural redesign of ETL pipeline to support petabyte-scale data ingestion, improving throughput by 560% and reducing processing time from weeks to hours
- Built query optimization engine leveraging cost-based planning, accelerating customer analytic queries by average of 3.2x across 50+ enterprise deployments
- Mentored 4 new engineers in distributed systems concepts; improved code review standards, reducing production bugs by 53%

PROJECTS

Meridian - Distributed Query Engine | *Rust, C++, gRPC, Protocol Buffers, Arrow* 2019 - Present (Open Source)

- Open-source columnar query engine achieving 4.8x performance improvement over Apache Spark for analytical workloads, with 3,200+ GitHub stars and 45+ active contributors
- Implemented adaptive query planning with ML-based cost estimation, improving query time prediction accuracy from 61% to 94%

CryptoWatch - Blockchain Event Monitor | *Go, PostgreSQL, Redis, Kafka, GraphQL* 2017 - 2019

- Engineered real-time blockchain transaction monitoring system processing 450K events/second with 99.99% uptime across 15 global nodes
- Reduced false-positive alerts by 89% through implementation of machine learning anomaly detection model trained on 2 years of historical data

EXTRACURRICULAR

Systems Research and Engineering Community (SREC) | *Founding Organizer & Speaker* 2019 - Present

- Founded and grew distributed systems meetup to 1,200+ active members across 3 cities, hosting 18+ technical talks annually from industry leaders
- Delivered 5 keynote presentations on distributed consensus and fault tolerance to audiences of 200+ engineers

- Contributed 340+ commits to high-profile open source projects (LLVM, Kubernetes, Protocol Buffers) with focus on performance optimization
- Mentored 8 junior developers entering tech industry, with 100% placement rate at tier-1 companies; designed curriculum for distributed systems fundamentals

TECHNICAL SKILLS

Languages – Rust, Go, C++, Java, Python, TypeScript, SQL

Technologies – Distributed Systems, Microservices Architecture, CRDT/Real-time Collaboration, gRPC, Kafka, Elasticsearch, PostgreSQL, Redis, GraphQL, Machine Learning Systems, Stream Processing

Tools – Kubernetes, Docker, Prometheus, Grafana, Protocol Buffers, Apache Arrow, Git, CI/CD Pipelines (GitHub Actions, Jenkins), AWS (EC2, S3, RDS), GCP (BigQuery, Bigtable)